
ReWAM: RECIPROCAL WORLD ACTION MODELS FOR INTERACTIVE AUTONOMOUS DRIVING

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Code: <https://github.com/LeapWM/rewam>

ABSTRACT

In interactive scenarios, an autonomous driving system is required to generate ego actions under the influence of other agents' behaviors. Existing World Action Models (WAMs) typically model other agents as components of the world model rather than as decision-makers that fundamentally shape the action of the ego agent, which impairs their performance in dense interaction scenarios. We introduce Reciprocal World Action Models (**ReWAM**), a game-theoretic world action modeling framework that captures the reciprocal influence between the ego agent and other agents by representing them as conditional responders whose actions are mutually influenced. We instantiate this framework with a Level- k response hierarchy, where role-specific ego and other action DiTs exchange compact strategy tokens through cross-agent attention while remaining grounded in a shared representation of the future driving world. To learn the response policy of the ego agent from demonstrations, we formulate expert actions as samples from the best response distribution and jointly optimize the entire hierarchy using conditional flow matching. Our framework is evaluated on the NAVSIM dataset and achieves state-of-the-art performance compared to baselines. The improvement is particularly significant in interactive scenarios, validating that modeling reciprocal responses provides a more effective foundation for interaction-aware world action generation.

1 Introduction

Autonomous driving requires an intelligent system to perceive complex traffic scenarios, anticipate their evolution, and plan a safe policy while interacting with multiple road users whose decisions are mutually dependent. End-to-end (Wang et al., 2026b; Guo et al., 2026; Xing et al., 2025; Jia et al., 2025b; Shang et al., 2026) autonomous driving simplifies conventional modular pipelines by jointly learning driving representations and action directly from historical sensor observations. Recent methods have improved this paradigm through multimodal feature fusion, goal-conditioned action generation, unified task modeling, and preference-aligned policy learning. However, the end-to-end paradigm primarily learns a direct mapping from observations to action, making it effective at reproducing demonstrated behavior, but action supervision alone does not ensure the policy acquires an explicit understanding of the environment and the mechanisms governing its evolution, thereby limiting its ability to generalize to real-world scenarios that fall outside the training distribution.

Compared with direct observation-to-action policies, Vision-Language-Action (VLA) models introduce language as a semantic interpreter between visual perception and action generation. By aligning visual observations with linguistic concepts, they are capable of interpreting road semantics, driving rules, and traffic participants' behaviors, and reasoning about an appropriate driving decision (Jiang et al., 2025; Kim et al., 2024; Wang et al., 2025; Zhou et al., 2026a; Liu et al., 2025; Zhou et al., 2026c). Reasoning-centric extensions further employ structured reasoning, cognitive feedback, and self-improvement to connect language-based scene understanding with executable action (Liu et al., 2026; Xiong

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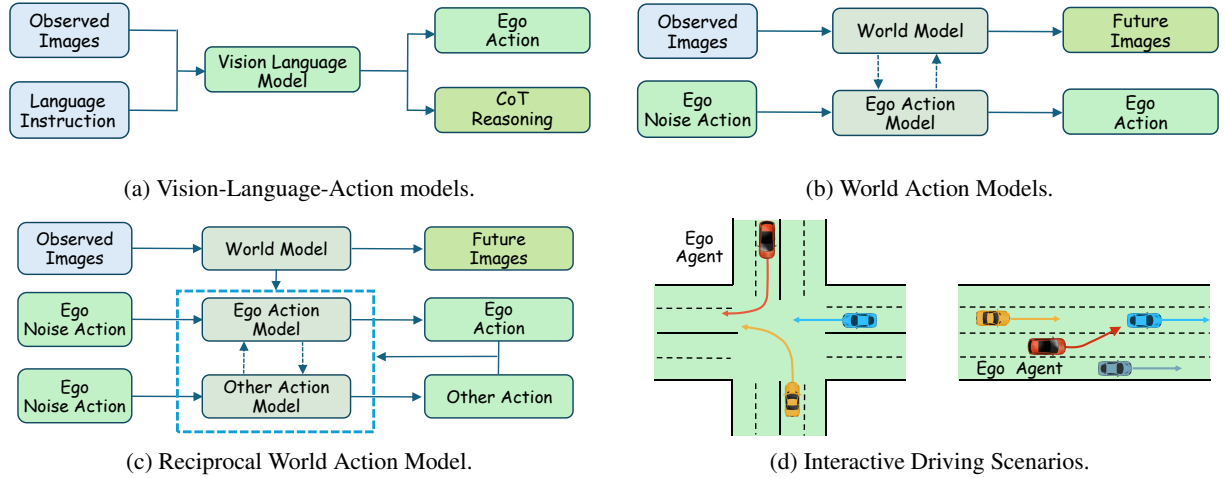


Figure 1: Comparison of different autonomous driving paradigms and representative interactive driving scenarios. (a) Vision-Language-Action models incorporate language-aligned semantic reasoning into action generation. (b) World Action Models ground ego-action generation in predicted world representations. (c) The proposed Reciprocal World Action Model explicitly captures reciprocal behavioral influence through recursive strategic reasoning. (d) Representative interactive scenarios in which the behaviors of the ego agent and other agents are mutually influenced.

et al., 2026; Zhang et al., 2026; Xiao et al., 2026). More recent methods combine VLA policies with prospective scene prediction or latent world modeling, allowing decisions to exploit imagined futures (Cen et al., 2025; Jia et al., 2026; Wang et al., 2026a; Fan et al., 2026; Xie et al., 2026; Chen et al., 2026a). These advances substantially strengthen semantic scene understanding and high-level reasoning. However, reasoning in the language space alone does not explicitly model the continuous physical evolution of the environment under different actions. A semantically plausible decision may therefore lack a grounded prediction of its consequences in the future world.

World models address a complementary limitation by learning how driving scenarios evolve over time. They forecast future observations or latent states conditioned on scenario history and candidate action, enabling a policy to evaluate possible consequences before acting (Zhou et al., 2026b; Zheng et al., 2025; Peng et al., 2026; Ye et al., 2026b). World Action Models (WAMs) further couple future generation and action prediction within a shared architecture, exposing predictive world representations directly to the planning process (Xia et al., 2026; Chen et al., 2026b; Wang et al., 2026c; Yuan et al., 2026; Ye et al., 2026a; Maes et al., 2026). Many predictive-representation approaches have demonstrated the value of compact latent dynamics for action-conditioned reasoning (Yang et al., 2026; Sun et al., 2026; LeCun et al., 2022). Despite this progress, existing WAMs primarily model how the ego action influences the future world. Other agents are usually predicted as part of that world, but their actions are not explicitly organized as strategic responses to the ego plan. Consequently, an action-conditioned future can be visually plausible without being behaviorally consistent with the interaction induced by other agents.

Interaction-aware prediction and planning methods explicitly model this reciprocal influence through conditional trajectory prediction, joint planning, or iterative behavioral refinement (Huang et al., 2023; Jia et al., 2025a; Tan et al., 2026; Li et al., 2025a; Liang et al., 2025). However, these methods predominantly operate in explicit state, maneuver, or action spaces, and their interaction modules are often separated from generative world modeling. This reveals a gap between two complementary directions: WAMs learn expressive action-conditioned representations of future driving worlds but lack an explicit strategic reasoning structure, whereas interaction-aware prediction and planning model behavioral responses but do not directly exploit the latent representations learned by generative world models.

Bridging these two directions requires more than adding an auxiliary other-agent prediction head to a WAM. A suitable interaction mechanism should preserve the learned representation of future world dynamics, represent ego and other agents' decisions as mutually dependent, and remain tractable in scenes involving multiple agents. In particular, it should capture a finite chain of behavioral responses without requiring the computation of a fully rational multi-agent equilibrium. A level- k game is a non-equilibrium model of bounded rationality that organizes agents into a hierarchy of strategic types, making it particularly well suited to autonomous driving. To date, the level- k game has been applied to lane changing, speed management, unsignalized intersections, and contingency planning (Karimi et al., 2023; Yuan et al., 2023; Chen et al., 2026c; Fang & Yi, 2024). However, existing formulations generally operate over handcrafted states, discrete maneuvers, or explicit actions. To the best of our knowledge, integrating the Level- k strategic hierarchy directly into the latent representations of a generative WAM remains unexplored.

In this paper, we propose **ReWAM**, a game-theoretic framework that incorporates bounded strategic reasoning into world-action modeling to explicitly capture how the ego agent and other agents mutually influence one another’s actions. Our core idea is to represent the ego agent and other agents as mutually responsive decision makers grounded in a shared future-world representation. At each reasoning level, the ego and other agents’ action branches synchronously update their action hypotheses by conditioning on the joint strategy predicted at the preceding level. This design retains the expressive latent dynamics of a WAM while making reciprocal behavioral responses explicit in action generation. Our contributions are summarized as follows:

- We explicitly model the reciprocal influence between ego and other agents’ actions within a WAM, extending WAMs beyond one-way ego-action conditioning toward interaction-aware action generation.
- We propose ReWAM, a game-theoretic world-action modeling framework that integrates bounded reasoning with role-specific action DiTs, enabling the consideration of the reciprocal influence between the ego and other agents’ actions.
- We evaluate the proposed ReWAM on NAVSIM, where it achieves state-of-the-art (SOTA) performance and delivers particularly pronounced improvements in scenarios with dense interactions, which establishes reciprocal response modeling as a critical capability for robust world-action generation in interactive scenarios.

2 Methodology

In this section, we introduce the proposed ReWAM framework, a game-theoretic WAM that augments generative world grounding with finite-depth strategic reasoning for interactive autonomous driving. Rather than treating the behavior of other agents as an implicit component of future-world prediction, ReWAM casts ego and other agents’ actions as mutually conditioned responses within a shared latent representation of the world. As illustrated in Figure 2, we formalize this hierarchy as a finite Level- k game, instantiate its response policies with separate ego and other-agent action models grounded in a shared world model, and learn the resulting conditional responses from expert demonstrations.

2.1 Level- k Game for Solving Interactive Policy

Consider a scenario containing an ego agent e and N other agents, with participant set $\mathcal{P} = \{e, 1, \dots, N\}$. At time t , the model receives an observation $\mathcal{O}_t$ containing the visual history, textual navigation condition, and current participant states s_t . Let τ_i denote the future action of participant i , and let $\tau_{-i} = (\tau_j)_{j \in \mathcal{P} \setminus \{i\}}$ indicate the joint action of all remaining participants. Similarly, π_i denotes the policy of participant i , and π_{-i} represents the corresponding joint policy of the remaining participants. Given π_{-i} , the expected return and best-response set of participant i are defined as

$$J_i(\pi_i, \pi_{-i}; \mathcal{O}_t) = \mathbb{E}_{\tau \sim (\pi_i, \pi_{-i})} [R_i(\tau; \mathcal{O}_t)], \quad (1)$$

$$\text{BR}_i(\pi_{-i}; \mathcal{O}_t) = \arg \max_{\pi_i} J_i(\pi_i, \pi_{-i}; \mathcal{O}_t), \quad (2)$$

where $\tau = (\tau_i, \tau_{-i})$ is the joint action and R_i is the reward of participant i . Directly coupling all participants at the same reasoning level produces circular dependencies and would generally require solving for a joint equilibrium, which is computationally prohibitive for real-time autonomous driving. Level- k game instead replaces such simultaneous coupling with a hierarchy of bounded strategic responses:

$$\begin{aligned} \pi_i^{(k)} &\in \text{BR}_i \left(\pi_{-i}^{(k-1)}; \mathcal{O}_t \right), \\ \text{BR}_i \left(\pi_{-i}^{(k-1)}; \mathcal{O}_t \right) &= \arg \max_{\pi_i} \mathbb{E}_{\tau \sim (\pi_i, \pi_{-i}^{(k-1)})} [R_i(\tau; \mathcal{O}_t)], \quad k \geq 1. \end{aligned} \quad (3)$$

where $\pi_i^{(k)}$ denotes participant i ’s policy at reasoning level k . This formulation converts simultaneous mutual dependence into a finite sequence of strategic responses, preserving reciprocal interaction while avoiding the computation of a joint equilibrium.

2.2 World Representation Conditioned Level- k Reasoning

As illustrated in Figure 2, ReWAM embeds the strategic hierarchy described in equation 3 into a generative WAM architecture, where a shared world model provides predictive context and dedicated Action DiTs parameterize the response policies of the ego agent and other agents. The world model utilizes the visual observations and textual navigation condition to generate a shared latent representation of the future driving world:

$$\mathbf{W}_t = f_\phi^{\text{WM}}(I_{t-H:t}, c_t) \in \mathbb{R}^{L_v \times d_v}, \quad (4)$$

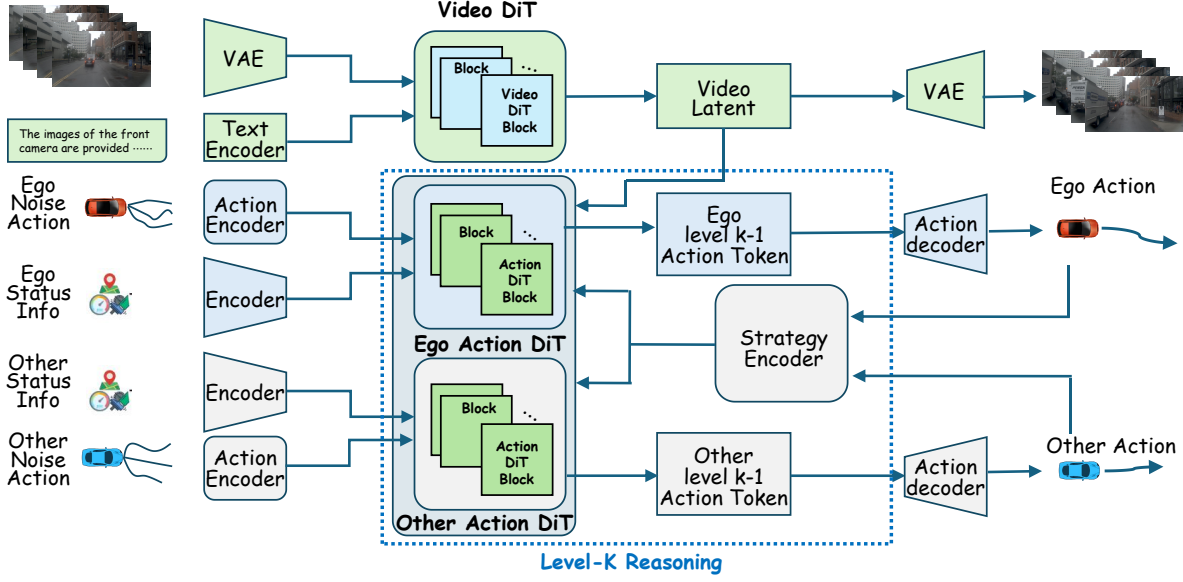


Figure 2: The proposed ReWAM framework. Visual observations and language instructions are processed by the video branch to construct a shared latent representation of the future driving world. Conditioned on this representation and agent-specific state information, the Ego and Other Action DiTs synchronously generate their action hypotheses. At reasoning level k , each branch conditions on the level $k - 1$ strategies of all remaining valid participants, forming a finite hierarchy of reciprocal responses.

where $I_{t-H:t}$ denotes the sequence of historical visual observations up to time t , c_t denotes the textual navigation condition, and L_v and d_v denote the number and dimension of world tokens, respectively. The same world tokens $\mathbf{W}_t$ are provided to all action branches so that their responses are grounded in a consistent representation of the driving world.

Grounded in the shared world representation, ReWAM employs two role-specific response generators: an Ego Action DiT parameterized by θ_e , which predicts the ego agent’s response to the preceding-level strategies of all other agents, and an Other Action DiT parameterized by θ_o , which is shared across other agents and predicts each other agent’s response. Let $\mathbf{Z}_{-i}^{(k-1)}$ denote the strategy encoder representation of the level $k - 1$ action of all participants other than i . Conditioned on a realized preceding-level strategy, the current action hypotheses are sampled as

$$\tau_e^{(k)} \sim q_{\theta_e}(\cdot \mid \mathbf{W}_t, s_t^e, \mathbf{Z}_{-e}^{(k-1)}), \quad (5)$$

$$\tau_i^{(k)} \sim q_{\theta_o}(\cdot \mid \mathbf{W}_t, s_t^i, \mathbf{Z}_{-i}^{(k-1)}), \quad i = 1, \dots, N \quad (6)$$

where $(\mathbf{W}_t, s_t^i)$ represents the decision-relevant information in $\mathcal{O}_t$ for participant i . Therefore, equation 5 and equation 6 provide amortized, action-conditioned realizations of the bounded responses in equation 3, where every level k prediction responds to a fixed realization of the remaining participants’ level $k - 1$ strategies, rather than solving a simultaneous equilibrium online.

2.3 Learning Best Responses from Demonstrations

Since specifying driving rewards for real-world scenarios is rather challenging, it is difficult to directly compute the best-response policies characterized in equation 2. We instead assume that the demonstrated action is generated by a joint expert policy $\pi^E = (\pi_j^E)_{j \in \mathcal{P}}$, whose components are reward-maximizing responses to the remaining expert policies:

$$\pi_i^E \in \text{BR}_i(\pi_{-i}^E; \mathcal{O}_t), \quad (7)$$

$$\tau^{\text{demo}} \sim \pi^E(\cdot \mid \mathcal{O}_t), \quad (8)$$

where τ^{demo} is the demonstrated joint action. Let $p_{\text{demo}}(\tau_i \mid \mathcal{O}_t, \tau_{-i})$ denote the conditional response distribution induced by the joint demonstrations. After applying the world, state, and strategy encoders, we use the same notation

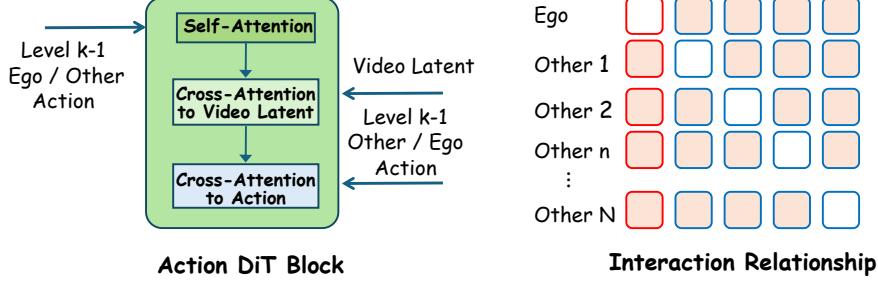


Figure 3: Architecture of the proposed Level- k interaction block. Within each Action DiT block, temporal self-attention models the target agent’s action sequence, while cross-attention to the Video DiT features grounds action generation in the predicted driving world. Cross-agent attention further conditions the current response on the preceding-level action hypotheses of the remaining agents. The elaborated Action DiT architecture explicitly captures reciprocal strategic influence among agents in interactive scenarios.

for the corresponding model-level conditional distribution $p_{\text{demo}}(\tau_i \mid \mathbf{W}_t, s_t^i, \mathbf{Z}_{-i})$. The response model can be learned through conditional distribution matching:

$$\theta_i^* = \arg \min_{\theta_i} \mathcal{D}(p_{\text{demo}}(\tau_i \mid \mathbf{W}_t, s_t^i, \mathbf{Z}_{-i}), q_{\theta_i}(\tau_i \mid \mathbf{W}_t, s_t^i, \mathbf{Z}_{-i})), \quad (9)$$

where $\theta_i = \theta_e$ for the ego vehicle and $\theta_i = \theta_o$ for other agents, respectively; and $\mathcal{D}$ measures the discrepancy between the demonstrated and learned conditional distributions. The resulting q_{θ_i} is therefore an amortized, stochastic surrogate for the best-response policy over the support of the demonstrated strategy distribution.

2.4 Level- k Interaction Block

As illustrated in Figure 2, ReWAM parameterizes the response distributions of the ego agent and other agents using two Action DiTs that progressively transform noisy action sequences into world-grounded and interaction-conditioned action predictions through stacked transformer blocks. Each block successively applies temporal self-attention over the target participant’s future action tokens, and cross-attention to the corresponding Video DiT features, cross-agent attention to the preceding-level action hypotheses, and a feed-forward transformation. This architecture integrates temporal action modeling, world-grounded contextualization, and explicit strategic interaction within one WAM framework. As shown in Figure 3, to model the interaction among agents, the cross-attention for participant i at transformer block ℓ is computed as

$$\mathbf{A}_{i,\ell}^{(k)} = \text{softmax} \left(\frac{Q_\ell(\mathbf{H}_{i,\ell}^{(k)}) K_\ell(\mathbf{Z}^{(k-1)})^\top}{\sqrt{d_h}} + \mathbf{M}_i \right) V_\ell(\mathbf{Z}^{(k-1)}), \quad (10)$$

where $\mathbf{H}_{i,\ell}^{(k)} \in \mathbb{R}^{T \times d}$ denotes the action tokens of participant i ; d_h is the attention-head dimension; the target participant’s action tokens serve as queries Q_ℓ , while the strategy representations of the preceding-level action provide keys K_ℓ and values V_ℓ . The mask $\mathbf{M}_i$ imposes role-dependent visibility by suppressing padded participants and the target participant’s own strategy token. Therefore, the proposed interaction blocks enable cross-agent attention to explicitly model the mutual influence among participants by conditioning each participant’s current response on the preceding-level action hypotheses of all other valid participants.

2.5 Joint Learning and Inference

Flow matching provides a tractable velocity-regression formulation of the distribution matching objective, so it is adopted to learn the response model targeted by equation 9. For participant $j \in \mathcal{P}$ at reasoning level k , let $\mathbf{x}_{0,j}$ denote a normalized demonstrated action sampled from the target conditional distribution:

$$\mathbf{x}_{0,j} \sim p_{\text{demo}}(\cdot \mid \mathbf{W}_t, s_t^j, \mathbf{Z}_{-j}^{(k-1)}). \quad (11)$$

Then, we sample $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $r \sim \mathcal{U}(0, 1)$, and define

$$\mathbf{x}_{r,j} = (1 - r)\mathbf{x}_{0,j} + r\epsilon, \quad (12)$$

$$\mathbf{u}_{r,j} = \epsilon - \mathbf{x}_{0,j}. \quad (13)$$

Table 1: Comparison with state-of-the-art methods on the NAVSIM benchmark. Higher values indicate better performance.

Method	Ref	NC↑	DAC↑	TTC↑	Comf.↑	EP↑	PDMS↑
<i>Traditional End-to-End Methods</i>							
VADv2-V ₈₁₉₂ (Jiang et al., 2024)	arXiv’24	97.2	89.1	91.6	100	76.0	80.9
UniAD (Hu et al., 2023)	CVPR’23	97.8	91.9	92.9	100	78.8	83.4
TransFuser (Chitta et al., 2023)	TPAMI’23	97.7	92.8	92.8	100	79.2	84.0
PARA-Drive (Weng et al., 2024)	CVPR’24	97.9	92.4	93.0	99.8	79.3	84.0
DiffusionDrive (Liao et al., 2025)	CVPR’25	98.2	96.2	94.7	100	82.2	88.1
ReCogDrive-IL (Xiong et al., 2026)	ICLR’26	98.1	94.7	94.2	100	80.9	86.5
<i>World Model Methods</i>							
DrivingGPT (Chen et al., 2025)	ICCV’25	98.9	90.7	94.9	95.6	79.7	82.4
LAW (Li et al., 2025b)	ICLR’25	96.4	95.4	88.7	99.9	81.7	84.6
Epona (Zhang et al., 2025)	ICCV’25	97.9	95.1	93.8	99.9	80.4	86.2
WoTE (Li et al., 2025c)	ICCV’25	98.5	96.8	94.9	99.9	81.9	88.3
DriveVLA-W0 (Li et al., 2026)	ICLR’26	98.4	95.3	95.2	100	80.9	87.2
PWM (Zhao et al., 2025)	NeurIPS’25	98.6	95.9	95.4	100	81.8	88.1
WorldDrive (Gui et al., 2026)	arXiv’26	98.4	96.8	95.2	100	83.3	89.0
DriveLaW (Xia et al., 2026)	CVPR’26	99.0	97.1	96.7	100	81.3	89.1
ReWorld (Peng et al., 2026)	arXiv’26	99.1	98.2	97.7	100	82.0	90.4
ReWAM (Ours)	-	99.3	97.8	97.2	100	83.6	90.6

where ϵ indicates a Gaussian noise sample that defines the source distribution; r is the continuous flow time that interpolates between the demonstrated action and noise; $\mathbf{x}_{r,j}$ denotes the action state at flow time r along the linear interpolation between the demonstrated action and Gaussian noise; $\mathbf{u}_{r,j}$ represents the corresponding target velocity. The resulting state-velocity pairs provide the regression targets for the conditional velocity field, which is learned by minimizing

$$\mathcal{L}_j^{(k)} = \mathbb{E} \left[\left\| v_{\theta_j} \left(\mathbf{x}_{r,j}, r \mid \mathbf{W}_t, s_t^j, \mathbf{Z}_{-j}^{(k-1)} \right) - \mathbf{u}_{r,j} \right\|_2^2 \right]. \quad (14)$$

Equation 14 defines the participant-level flow-matching objective for a single response at reasoning level k , and these objectives are subsequently aggregated according to the two response roles:

$$\mathcal{L}_{\text{ego}}^{(k)} = \mathcal{L}_j^{(k)} \Big|_{j=e}, \quad \mathcal{L}_{\text{other}}^{(k)} = \frac{\sum_{i=1}^N m_i \mathcal{L}_i^{(k)}}{\sum_{i=1}^N m_i}, \quad (15)$$

where $m_i \in \{0, 1\}$ indicates the validity of surrounding agent i . The branch-level objectives are aggregated across the reasoning hierarchy using normalized depth-dependent weights:

$$\mathcal{L}_{\text{flow}} = \sum_{k=1}^{K_{\text{train}}} \alpha_k \left(\lambda_{\text{ego}} \mathcal{L}_{\text{ego}}^{(k)} + \lambda_{\text{other}} \mathcal{L}_{\text{other}}^{(k)} \right), \quad \alpha_k = \frac{k^p}{\sum_{\ell=1}^{K_{\text{train}}} \ell^p}. \quad (16)$$

where K_{train} specifies the supervised reasoning depth; α_k denotes the normalized depth-dependent weight; p controls the relative emphasis assigned to deeper response levels; λ_{ego} and λ_{other} balance the two action branches; the normalization ensures $\sum_{k=1}^{K_{\text{train}}} \alpha_k = 1$. Accordingly, the response models are jointly optimized under the aggregate objective:

$$\Theta^* \in \arg \min_{\Theta} \mathcal{L}_{\text{flow}}(\Theta), \quad (17)$$

where Θ denotes the complete set of trainable response-model parameters. At the population optimum, conditional flow matching recovers the demonstration-conditioned response distribution for each valid participant and supervised reasoning level, which means

$$q_{\theta_j^*} \left(\cdot \mid \mathbf{W}_t, s_t^j, \mathbf{Z}_{-j}^{(k-1)} \right) = p_{\text{demo}} \left(\cdot \mid \mathbf{W}_t, s_t^j, \mathbf{Z}_{-j}^{(k-1)} \right), \quad j \in \mathcal{P}, \quad k = 1, \dots, K_{\text{train}}. \quad (18)$$

where $\theta_j^* = \theta_e^*$ for the ego agent and $\theta_j^* = \theta_o^*$ for other agents. Therefore, the distribution-matching objective in equation 9 is achieved across response roles and reasoning levels by conditional flow matching.

3 Experiments

3.1 Experimental Setup

Implementation details. We implement the proposed ReWAM on top of DriveLaW (Xia et al., 2026) by designing the Level- k interaction block and introducing a parameter-shared Other Action DiT together with a strategy encoder. The pretrained VAE, text encoder, and Video DiT are frozen, while both Action DiTs and the associated interaction modules are optimized. We first train the Level-1 model and initialize every $k \geq 2$ model from the resulting checkpoint for further fine-tuning. We use four front-camera frames sampled at 2 Hz and predict eight poses at 0.5 s intervals for the ego agent and at most ten other agents within 80 m. Current agent states are obtained from scene annotations, whereas future agent states and actions are used only as supervision.

Dataset and evaluation metrics. We conduct experiments on NAVSIM, a planning-oriented benchmark derived from the public OpenScene/nuPlan driving logs. We use the official `navtrain` split for training and the `navtest` split for evaluation. The latter contains 136 logs and 12,146 driving scenarios. Unless stated otherwise, we report the Predictive Driver Model Score (PDMS) together with its components: No-at-Fault Collision (NC), Drivable Area Compliance (DAC), Time to Collision (TTC), Comfort, and Ego Progress (EP). We additionally report Driving Direction Compliance (DDC) as a diagnostic measure. All metric values are multiplied by 100, and higher values indicate better performance.

3.2 Main Results

Comparison with SOTA methods. As presented in Table 1, ReWAM achieves the highest PDMS score of 90.6, improving over its DriveLaW backbone by 1.5 points and the SOTA method, ReWorld, by 0.2 points. The gain over DriveLaW is accompanied by improvements in NC, DAC, TTC, and EP scores, with the largest enhancement appearing in EP score. Although ReWorld obtains slightly higher DAC and TTC scores, ReWAM achieves better performance in NC and EP scores and consequently a better aggregate score. These results validate that explicitly modeling the interaction between the ego agent and other agents enables the planner to better anticipate reciprocal behavioral responses, thereby achieving a SOTA performance on the benchmark.

3.3 Analysis of Level- k Reasoning

Effect of reasoning depth We vary the reasoning depth while keeping the model and evaluation protocol fixed, and report the performance at each reasoning level in Table 2. Increasing the depth from Level-1 to Level-4 consistently improves the aggregate performance, raising PDMS from 90.03 to 90.64. The largest single-level gain occurs at Level-2, whereas subsequent levels deliver progressively smaller improvements. This trend is most pronounced in TTC, DAC, and EP, indicating that recursive response reasoning enhances collision avoidance, drivable-area compliance, and driving progress, while Comfort and DDC remain consistently high across all depths. Although Level-5 marginally improves NC and EP, it yields no further gain in PDMS and slightly reduces DAC. These results suggest that deeper reasoning progressively captures strategic dependencies among traffic participants, but exhibits diminishing returns once the dominant interaction structure has been resolved.

Benefits across interaction intensities. To assess whether Level- k reasoning yields greater benefits in interactive settings, we partition all valid tokens into five interaction-intensity groups (None, Weak, Medium, Strong, and Critical) using a model-independent score derived from the official reference trajectories. As shown in Table 3, ReWAM consistently outperforms both baselines across all five groups, with its advantage becoming particularly pronounced in the most interaction-intensive scenarios. Specifically, its improvement over DriveLaW increases from 1.52 PDMS scores in the None group to 4.59 in the Critical group, where it also surpasses DiffusionDrive by 2.54. Such concentration of performance gains in challenging interactive scenarios provides strong empirical support for the effectiveness of explicitly modeling reciprocal responses among traffic participants.

Performance decomposition under dense interactions. We further examine the Critical subset, and the results are shown in Table 4. ReWAM exhibits component-wise dominance over both baselines across all non-saturated metrics, while retaining the same perfect Comfort score. The concurrent improvements in NC and TTC indicate a stronger capacity to anticipate and resolve spatiotemporal motion conflicts. Importantly, these safety gains are accompanied by higher DAC, DDC, and EP, suggesting that conflict avoidance is achieved without resorting to off-road, directionally inconsistent, or excessively conservative behavior. The resulting PDMS improvement therefore reflects a more favorable coordination of safety, traffic-rule compliance, and driving progress, rather than optimization of any single metric. This consistent performance profile provides further evidence that explicit reciprocal-response reasoning is particularly effective when participants’ decisions are strongly interdependent.

Table 2: The performance of the proposed method across different reasoning levels.

Interaction Level	NC	DAC	TTC	Comfort	EP	DDC	PDMS
Level-1	99.06	97.39	96.25	99.97	83.28	98.12	90.03
Level-2	99.25	97.68	97.13	99.97	83.44	98.17	90.48
Level-3	99.26	97.74	97.13	99.98	83.54	98.15	90.55
Level-4	99.25	97.81	97.18	99.96	83.60	98.14	90.64
Level-5	99.29	97.77	97.18	99.97	83.64	98.16	90.64

Table 3: PDMS scores across interaction-intensity groups on NAVTEST. Share denotes the percentage of the 12,144 evaluated tokens, and parentheses report ReWAM’s absolute improvement over each baseline.

Interaction Intensity	Tokens	Share (%)	ReWAM	DriveLaW	DiffusionDrive
None	6621	54.52	89.92	88.40 (+1.52)	87.06 (+2.86)
Weak	1841	15.16	90.89	89.76 (+1.13)	89.20 (+1.69)
Medium	1841	15.16	92.33	91.73 (+0.60)	89.56 (+2.77)
Strong	1288	10.61	92.43	90.50 (+1.93)	90.17 (+2.26)
Critical	553	4.55	88.72	84.13 (+4.59)	86.18 (+2.54)

Table 4: Performance of the proposed method and baselines on critical-interaction scenarios.

Method	NC	DAC	TTC	Comfort	EP	DDC	PDMS
ReWAM	96.93	99.10	85.90	100	89.59	98.19	88.72
DriveLaW	94.21	98.55	79.20	100	87.06	98.01	84.13
DiffusionDrive	95.39	97.47	84.27	100	87.07	97.38	86.18

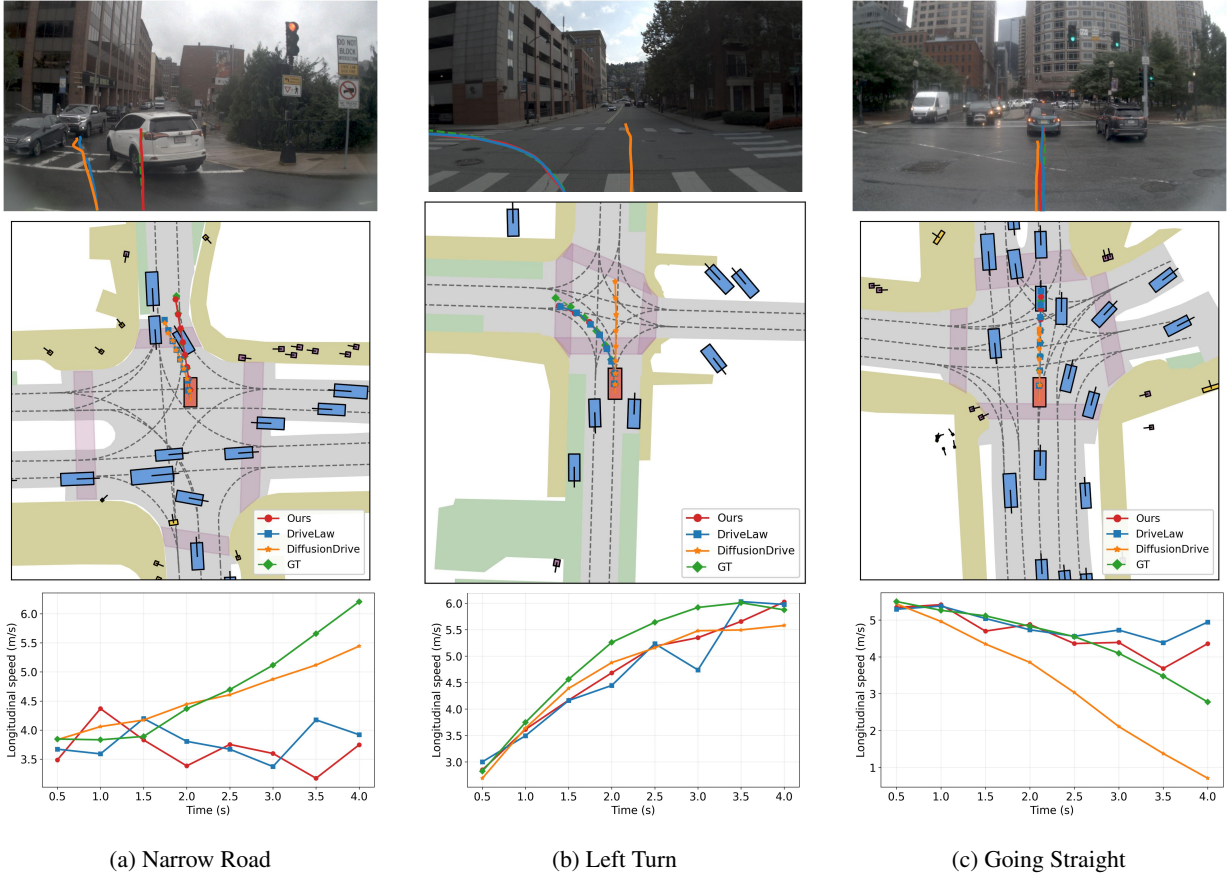


Figure 4: Qualitative comparison across three representative interactive driving scenarios. Each panel presents the front-camera action projection, bird’s-eye-view trajectory visualization, and longitudinal speed profile.

3.4 Qualitative Results

The results of three interactive scenarios, including front-camera-view action projection, bird’s-eye-view action visualization, and longitudinal speed profile, are shown in Figure 4. In the narrow road scenario, ReWAM successfully negotiates the constrained passage, whereas both baselines fail to complete the traversal. In the left turn scenario, DiffusionDrive maintains an almost straight trajectory and misses the intended maneuver, while ReWAM conforms to the turning geometry and preserves smooth progress. The going straight scenario further exposes two opposing failure modes: DiffusionDrive decelerates excessively and sacrifices ego progress, whereas DriveLaW responds insufficiently and collides with the leading vehicle. By contrast, ReWAM adopts a properly calibrated trajectory and speed profile that avoids the conflict without unnecessary loss of progress. These examples collectively demonstrate the superior performance of ReWAM in interactive driving scenarios, where reciprocal response modeling yields safer and more effective action.

4 Conclusion

In this paper, we presented the proposed ReWAM, a game-theoretic world-action modeling framework that explicitly captures the reciprocal influence between the action of the ego agent and other agents, advancing WAMs from one-way ego-action conditioning toward strategically coupled multi-agent action generation. The proposed formulation replaces simultaneous multi-agent coupling with a finite hierarchy of bounded responses, thereby introducing structured strategic reasoning without incurring online equilibrium computation. Within this hierarchy, the Ego and parameter-shared Other Action DiTs are grounded in a shared future-world representation and exchange preceding-level action hypotheses through cross-agent attention. Conditional flow matching then learns the corresponding response distributions directly from demonstrations. ReWAM achieves the best performance on NAVSIM, with a PDMS of 90.6, while the study on reasoning depth shows that recursive updates provide consistent gains before saturating at deeper levels. Interaction-stratified evaluation further demonstrates that our method outperforms both DriveLaW and DiffusionDrive across all intensity groups, with its largest advantage over the DriveLaW backbone arising in the critical interaction group. The accompanying improvements in safety, traffic-rule compliance, and ego progress, together with the qualitative examples, validate that these gains arise from more effective interaction-aware action generation rather than conservative planning alone. These findings establish bounded reciprocal-response reasoning as a practical and effective foundation for interactive world-action modeling.

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